Folding a Ring onto Heavy-Hex: Formula-Built Schedules with Provable Gate Counts, and What Their Signal Shows on Twins and Devices

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Version 1.3 — 31 October 2026

Abstract

A ring of n CZ gates per layer, the entangling pattern of many variational classifiers, does not fit IBM's heavy-hex devices unless n is a cycle length of the lattice, and heavy-hex has cycles of $12, 20, 24, 28, \ldots$ qubits only. We compare three schedules that a compiler can build by formula, with no layout search and without reading the day's calibration: the *zipper* (the ring folded onto a line; $5n/2 - 4$ two-qubit gates per layer for even n, $(5n - 7)/2$ for odd n), *holes* (the ring on a longer cycle, its spare qubits moved as $|0\rangle$; $n + 2(m - n)$), and *pendants* (the ring on a shorter cycle, its extra nodes hung off it; $n + (n - m)$, on exactly n qubits). We prove that the pendant schedule's count is the least possible for layered schedules on the cycle and its leaves in a stated routing model (exhaustive search finds the same for interleaved ones in small cases), that the hole schedule's count is least within its class but not in general, and recall that no cheaper schedule than the zipper was found by exhaustive searches at $n = 4$–12 and 14. A rule that takes the fewest extra gates picks pendants at $n = 13$–17 and holes at $n = 8$–11 and 18–19 on IBM's Heron devices; the rule is in PSF-Zero's AI front end a26. On noise models, in declared matches against the zipper at 16 layers over four random classifiers, the rule won at every $n = 13$–18 on two twins. Its lead was 1.5–8.2× the zipper's kept signal on the twin of a 13-day-old snapshot and only 1.07–1.28× (about even at $n = 18$) on the twin of a real day's calibration, the kind that had predicted the zipper on hardware: the old snapshot was pessimistic for the zipper. On two devices (`ibm_kingston`, `ibm_fez`) the pendant schedule kept the signal its day's twin predicted (0.90 ± 0.12 against 0.77; 0.50 ± 0.10 against 0.59), in two jobs of 7 s each; these are coarse tests and no device comparison with the zipper was made. Against Qiskit's search (level 2 on the same twin) the rule won on every classifier at every n simulated ($n = 13$–16, 18): Qiskit's circuits kept −0.05–0.64 of the signal against 0.61–0.82. The bare ring kept 0.78–0.86 at $n = 20$ and 0.81–0.83 at $n = 24$. On one 20-qubit cycle, two pendants kept nearly what the bare ring kept (0.75–0.84 against 0.78–0.86) and clearly more than two holes (0.61–0.68), with different n and classifiers; in a separate test with the classifiers and qubits fixed and only the schedule changed, the bare ring kept 1.44–1.81 times the zipper's signal and two holes 1.13–1.20 times. Error-product estimates of the kept signal did not rank the schedules. We give the benchmark number and the device user's number side by side throughout.

1 Introduction

Variational quantum classifiers often entangle their qubits with a ring of CZ gates in every layer [6]. On a device whose coupling graph contains the ring, the layer costs n CZ. IBM's Heron processors are heavy-hex lattices [5], which have no cycle shorter than 12 qubits and no cycle between 12 and

20 (Section 2). Every other ring size needs routing. A search-based compiler places and routes the ring with the day's calibration; PSF-Zero [1] builds what it can by formula instead and refuses the rest. This paper is about the formula side: which ring schedules can be written down from the coupling graph alone, how many two-qubit gates they need, whether that is the least, and how much signal they keep.

Our earlier paper [2] compared the zipper with Qiskit's search [4] through a trained classifier and found them indistinguishable on the declared tests. Here we add two schedules that use the device's own cycles, a count rule that chooses among the three, proofs or exhaustive checks of their counts, matches on noise models declared before the runs, and the first device runs of the new schedule. All runs, declarations and records are in TOQB [3]; the compiler is PSF-Zero's AI front end a26.

Related work and what is new here. Laying the qubits of a structured circuit along a path of the device and swapping qubits off that path in and out is known: IBM's swap strategies for QAOA on heavy-hex use the longest line and periodic swaps with qubits beside it [9], and a linear-depth QFT has been mapped onto a main line with qubits hung off it [10]. Merging a CZ with the SWAP that follows it, and swapping with a qubit known to be in $|0\rangle$ by two CX, are known building blocks [20, 11]. That compiled circuits are ranked only imperfectly by their two-qubit gate count or by error-product estimates has been shown on hardware [12, 13], and that a schedule's timing changes the error on the same qubits and gates [14]. The slope of the noisy on the noiseless expectation value is the rescaling factor of error mitigation [15, 16], and noise outside an observable's backward light cone does not affect it [17]. How transpiling a ring ansatz onto heavy-hex changes its expressibility and gradient variance has been studied in noiseless simulation [18]; we study the signal kept under noise, with exact gate counts. What we add is narrower: the ring problem on heavy-hex cycles, the exact per-layer counts of three schedules, with lower bounds in stated models for two of them and exhaustive searches for the third; the kept-signal slope as the measure of schedules; one test that holds the classifier, the inputs and the qubits fixed and changes only the schedule; and every match declared, with its decision rule, before it ran. Our "folding" is a placement of the ring on a cycle; it is unrelated to the gate folding of zero-noise extrapolation [19].

2 Heavy-hex cycles

Heavy-hex is the honeycomb lattice with every edge subdivided by one qubit [5]. A cycle in it is a cycle of the honeycomb with every edge subdivided, which doubles its length. Honeycomb cycles have length 6 (one hexagon) and every even length from 10 on (outlines of clusters of hexagons: two adjacent, 10; three around a vertex, 12; three in a row, 14; four with one inner vertex, 16; ...), and, the lattice being bipartite with girth 6, none of length 8. Heavy-hex cycles therefore have lengths $12, 20, 24, 28, \ldots$, with none between 12 and 20. We checked this on the coupling graphs of four Heron fake devices by listing every simple cycle up to length 22: only 12 and 20 occur (21 and 44 cycles on `FakeMarrakesh`, `FakeFez` and `FakeKingston`; 18 and 37 on `FakeTorino`).

A ring of n embeds without extra gates only when n is one of these lengths and a working cycle of that length exists. Every other n needs extra two-qubit gates.

3 Three schedules built by formula

Placement. The lattice is read off the coupling graph: *bridges* (qubits with two neighbours whose indices are not consecutive), *rows* (the components left without them) and *cells* (between

consecutive bridges of a row pair). A cycle is the outline of a fixed shape of cells: k cells side by side in a row pair ($8k+4$ qubits: 12, 20, 28), or k cells plus k_2 cells in the next row pair down (24, 32, 36, ...). The first shape in a fixed order (row pairs top to bottom, cells left to right) whose qubits and couplers all work is used. No search over mappings and no error value decides; failed elements are never used. The compiler (a26) treats a qubit or coupler at error ≥ 0.5 as failed; TOQB's analysis arm, which built every circuit run in this paper, uses error ≥ 1. On the marrakesh snapshot and on the `ibm_kingston` Target of 24 October no element has an error in $[0.5, 1)$, so the two give the same circuits there (on the snapshot this was also checked instruction for instruction, $n = 8$–19).

Zipper. The ring laid on a line in folded order $(0, 1, 2, n-1, n-2, 3, 4, \dots)$; each layer a fixed sequence of CZ rounds, CZ+SWAP blocks at the line's ends and interior SWAPs; the next layer runs backwards. Two-qubit gates per layer: $5n/2 - 4$ (even n), $(5n-7)/2$ (odd n) [2, 3].

Holes. The ring on the shortest working cycle of $m \geq n$ qubits; the $h = m - n$ spare qubits start in $|0\rangle$ and are spread evenly. Each layer: CZ on every ring edge already adjacent; for each run of holes the ring node before it moves across (CX $a \to e$, CX $e \to a$ for each empty qubit e), then CZ across. Two-qubit gates per layer: $n + 2h$ when holes are apart ($n = 8$, where two holes meet, needs two more per layer in some layers).

Pendants. The ring on the longest working cycle of $m \leq n$ qubits; the $k = n - m$ extra ring nodes sit on *leaves*, outside neighbours of k cycle nodes (*hubs*), spread in cycle order. Each layer every leaf node trades places with its hub's node by one CZ+SWAP block, which also applies their ring edge. Two-qubit gates per layer: $n + k$, on exactly n qubits; no idle qubit is carried.

The count rule. Of the extra two-qubit gates per layer — pendants $n - m_{\leq}$, holes $2(m_{\geq} - n)$, the zipper's — the fewest decides (ties: pendants, then holes). Table 1 gives its choices on a Heron device.

n	8–11	12	13–17	18–19	20	21–22	23	25–26
choice	holes (12)	ring	pendants (12)	holes (20)	ring	pendants (20)	holes (24)	pendants (24)
gates per layer	$16, 15, 14, 13$	12	$14 \dots 22$	$22, 21$	20	$22, 24$	25	$26, 28$
zipper	$16, 19, 21, 24$	26	$29 \dots 39$	$41, 44$	46	$49, 51$	54	$59, 61$

Table 1: The count rule's choice (cycle length in brackets) on `snapshot:marrakesh`; at $n = 8$ the counts tie and the rule takes holes. Not shown: the ring itself at $n = 24, 28$, holes at $n = 27$ and pendants at $n = 29, 30$ (cycle 28). At $n = 20$–30 the chosen schedule uses 0.42–0.47 of the zipper's two-qubit gates.

Correctness. Every schedule equals the logical model on every measured bit: by statevector for $n \leq 19$ ($L = 2, 4$), by Clifford simulation with random multiples of $\pi/2$ for $n = 20$–30, and, for the pendant schedule, by a structural replay of every pendant placement on cycles up to $m = 12$ (8,184 cases) and a statevector check of every placement up to $m = 8$ (1,398 cases), done in a separate review that did not change the code. The compiler's circuits equal TOQB's analysis arm instruction for instruction ($n = 8$–19).

4 Gate-count lower bounds

Routing model. Each ring node's state sits on one qubit at a time (or on none, for an empty qubit in $|0\rangle$). A two-qubit operation either applies a ring-edge CZ between the nodes it holds, or changes where states sit, or both; no measurement, feed-forward, teleportation or extra entangled resource. In layer t every ring edge is applied once, after the single-qubit gates of layer t on both its nodes and before those of layer $t+1$. A schedule is *layered* if all operations of layer t precede those of layer $t+1$. Costs are counted in two-qubit gates; an operation's *extra* is its cost minus the ring edges it applies (at most one).

Fact 1. *A CZ followed by a SWAP on the same pair needs two CZ (it is locally equivalent to iSWAP [7, 8]); a SWAP needs three. A single CZ with single-qubit gates neither exchanges two states nor moves a state onto a qubit in* $|0\rangle$.

For the last claim: with local gates folded in, the CZ acts on $|\chi\rangle \otimes |\varphi\rangle$ and gives $\alpha|0\rangle|\varphi\rangle + \beta|1\rangle Z|\varphi\rangle$; if $Z|\varphi\rangle \propto |\varphi\rangle$ the state stays on the first qubit, otherwise the output is entangled for generic α, β. Hence an operation that moves states has extra at least 1 (CZ+SWAP, which also applies a ring edge), 2 (a move onto an empty qubit, two CX) or 3 (a SWAP). An operation that only exchanges two nodes is a SWAP: an exchange in two CZ is CZ+SWAP, which also applies a CZ between the two nodes, allowed only when it is their ring edge not yet applied in the layer.

Lemma 1 (pendants)**.** *Let the hardware be a cycle C_m plus k leaves, each attached to a distinct hub, no two hubs adjacent (so $k \le \lfloor m/2 \rfloor$), every qubit holding a ring node ($n = m + k$). In the routing model every layer of a layered schedule costs at least $n + k$ two-qubit gates. The pendant schedule is layered and meets the bound.*

Proof. Fix a layer and a leaf ℓ with hub h; let x be the node on ℓ when the layer starts. If x leaves ℓ during the layer, an operation that moves states acts on (h, ℓ), the leaf's only coupler. Otherwise both ring edges of x, with $x-1$ and $x+1$, are applied across (h, ℓ) in this layer, so h holds $x-1$ at one time and $x+1$ at another; since no qubit is empty and x stays on ℓ, the node on h changes through an operation that moves states on a cycle edge at h; it lies between two operations of this layer, so in a layered schedule it belongs to this layer. Name that operation ℓ's. Operations named for different leaves are different: one on a leaf edge belongs to one leaf, and a cycle edge at h is a cycle edge at another hub h' only if h and h' are adjacent, which is excluded. The layer thus holds at least k distinct state-moving operations, each with extra ≥ 1, besides the n ring edges; its cost is at least $n + k$. The pendant schedule finishes each layer before the next, with one CZ+SWAP (extra 1) per leaf and plain CZs otherwise. □

On heavy-hex every coupler joins a degree-3 qubit to a degree-2 qubit and a hub has degree 3, so no two hubs are adjacent; the coupling graphs of four Heron fake devices have no coupler between two degree-3 qubits.

Checks by search. A shortest-path search over layouts and applied ring edges, from every start layout, with the operations CZ (1), CZ+SWAP (2), SWAP (3) and move onto an empty qubit (2), found the cheapest single layer to be exactly $n + k$ on cycles $m = 4, 5, 6$ with every set of one or two hubs (up to rotation and reflection), adjacent hubs included. Over L layers with interleaving allowed (a ring edge may be applied for the $(c+1)$-th time once both neighbouring ring edges have been applied c times), the least total was exactly $L(n + k)$ in every case tried: one hub on $m = 4$ ($L = 2$–4), $m = 5$ ($L = 2, 3$) and $m = 6$ ($L = 2$); two non-adjacent hubs on $m = 4$ ($L = 2, 3$) and $m = 5$ ($L = 2$). We have not proved the bound for interleaved schedules in general.

Lemma 2 (holes, within their class)**.** *On a cycle C_m with h empty qubits and $n = m - h \geq 3$ nodes, among schedules in which nodes move only onto empty qubits (never exchange, so their cyclic order is kept), every layer of a layered schedule costs at least $n + 2h$.*

Proof. Since nodes never pass each other, two nodes that are not consecutive in the cyclic order never become adjacent; for every ring edge to be applicable, each node's two ring neighbours must therefore be its two cyclic neighbours ($n \geq 3$), and the cyclic order is the ring order or its reverse. Let q be a qubit empty at the start of a layer. If no node enters q during the layer, no node passes it, so the two nodes beside it stay the same ring-consecutive pair x, $x + 1$; they are separated by q on one arc and by the other $n - 2 \geq 1$ nodes on the other, so they are never adjacent and their ring edge cannot be applied. Hence each of the h qubits empty at the start is entered during the layer, by distinct operations (each enters one qubit) that cost 2 and apply no ring edge (a CZ with an empty qubit does nothing); the n ring edges need n more. □

The bound does not rest on excluding operations that spread a state over two qubits: bridging an edge across r empty qubits (CX, ..., CZ, ..., CX) costs $2r + 1$, again 2 extra per empty qubit, in every layer. The hole schedule meets the bound when no two holes meet (counted at $n = 9, 10, 18$: 15, 14, 22 per layer); at $n = 8$, where moved holes meet, it uses 4 more over 16 layers. Outside the class the bound fails when holes are many. The same search, for a single layer from the best start on a bare cycle, finds 6 for $m = 6, 7$ with $n = 4$ ($n + 2h = 8$, 10), 9 for $m = 8$, $n = 5$ (11) and 11 for $m = 9$, $n = 6$ (12), each by leaving the order (as the zipper does); with two holes on $m = 8$ it finds $n + 2h = 10$. Where the count rule picks holes on Heron, $h \leq 4$; whether $n + 2h$ is least there without the class restriction is open.

The zipper. Its count is a search result, not a proof: from every start layout for $n = 4$–10 (91,296 start layouts up to symmetry at $n = 10$, of which one reaches the least) and from every minimum-length start at $n = 12, 14$, no cheaper schedule was found; for odd n the same holds from every minimum-length start for $n = 5$–11 and from every start for $n = 5, 7, 9$ [3].

5 Signal on noise models

Measure. One random classifier of TOQB's ring family (single-qubit rotations per layer, a CZ ring, 16 layers, $\langle Z \rangle$ of qubit 0), 24 random inputs, 256 noise trajectories; the kept signal λ is the slope of the noisy $\langle Z \rangle$ on the noiseless one, $z \approx \lambda z_0 + \delta$ (the rescaling factor of error mitigation [15, 16]). Shots for a fixed precision scale as $1/\lambda^2$. Two twins: one built from `ibm_marrakesh`'s published calibration of 13 October 2026 (the *snapshot*), one from `ibm_kingston`'s calibration of 24 October 2026 (the *day twin*), which on that day had predicted the zipper on hardware (the device's slope on the twin's prediction 1.00, 0.95–1.02 per classifier) [2].

Reading. The direction held on both twins; the size did not. On the snapshot the zipper kept 0.09–0.46 and the rule won by 1.5–8.2×; on the day twin the zipper kept 0.58–0.77 and the lead was 1.07–1.28× (shots 1.15–1.6× fewer), about even at $n = 18$. The vanity number (half the two-qubit gates, a shorter duration) holds on both; the essential number is the day twin's.

Holes against the zipper where holes are many. On the snapshot, at $L = 16$: $n = 8$ (4 holes) the zipper kept more on every classifier (0.19–0.25 against 0.10–0.19), $n = 9, 10$ (3, 2 holes) the holes won on every classifier (0.26–0.37 and 0.51–0.56 against 0.07–0.17). At $n = 8$ the counts

n	13	14	15	16	17	18
rule's choice	pendants	pendants	pendants	pendants	pendants	holes
two-qubit gates, rule / zipper	224 / 464	256 / 496	288 / 544	320 / 576	352 / 624	352 / 656
snapshot: rule's λ	0.68–0.73	0.64–0.76	0.70–0.72	0.63–0.73	0.61–0.68	0.51–0.59
snapshot: zipper's λ	0.09–0.27	0.13–0.46	0.24–0.31	0.17–0.40	0.21–0.31	0.26–0.34
snapshot: verdict	WIN 4/4	WIN 4/4	WIN 4/4	WIN 4/4	WIN 4/4	WIN 4/4
day twin: rule's λ	0.76–0.81	0.73–0.82	0.78–0.81	0.77–0.80	0.69–0.75	0.61–0.68
day twin: zipper's λ	0.64–0.70	0.63–0.77	0.67–0.69	0.60–0.70	0.58–0.65	0.63–0.66
day twin: ratio	1.15–1.19	1.07–1.18	1.15–1.19	1.12–1.28	1.15–1.28	0.98–1.06
day twin: verdict	WIN 4/4	WIN 4/4	WIN 4/4	WIN 4/4	WIN 4/4	WIN 3/4

Table 2: Declared matches, $L = 16$, four random classifiers each (ranges over the four; on the snapshot both the twin and a perturbed stand-in were required). WIN: the rule's λ larger on at least 3 of 4 classifiers. Both predictions held.

tie; the rule's tie goes to holes and was left so. These are snapshot numbers, so the size of the $n = 9, 10$ leads is an upper bound.

Where the read-out node sits. Moving the read-out node off the pendants by formula gained 0.01–0.05 at $n = 14$–17 and lost 0.29–0.39 at $n = 13$, where the moved node landed next to the cycle's poorest qubit. A formula that reads no error values cannot avoid such places; it was not adopted.

Error products do not predict the signal. Two estimators without simulation, the product of $(1 - \text{error})$ over the whole circuit and over the read-out's backward light cone, ranked 64 simulated cells with Spearman correlation 0.45–0.68 and ordered 82–93 % of clearly different pairs; both failed the test we had fixed for using them (0.8 and 90 % at both depths). The noise reaching the read-out depends on the circuit's shape, not only on how many errors it contains.

Larger rings, one cycle. At $n = 20$ and 22 the rule's signal was simulated on the day twin too, rule only (declared; four models; the zipper would need 736 and 816 two-qubit gates). On this Target the rule uses the same 20-qubit cycle at $n = 18$, 20 and 22, so three placements can be compared on the same qubits (Table 3). Two pendants, with the most two-qubit gates, kept nearly what the bare ring kept and clearly more than two holes: on the same qubits the placement, not the gate count, decided most of the difference. The random classifiers differ between n, so the comparison mixes the placement with the classifier; the ranges of four classifiers do not overlap between the holes and the other two. At $n = 24$ (declared separately; the ring itself on a 24-cycle, 384 two-qubit gates, 5.6 μs) the rule kept 0.81–0.83 on four classifiers (prediction ≥ 0.61 held), the level of the bare ring on the 20-cycle: going from 20 to 24 qubits on a cycle cost no measurable signal here, though the place and the classifiers changed with n.

Same classifier, same qubits, two schedules. The comparison above mixes the placement with the classifier. A declared match removed that: at $n = 18$ and 20, the same four classifiers and inputs ran with the zipper laid along the same cycle's qubits (a path of 18 or 20 of them), against the rule's rows. The rule won on every classifier at both n (Table 4); the $n = 20$ prediction (WIN) held, no direction had been predicted at $n = 18$. With everything but the schedule fixed, the bare ring kept 1.44–1.81 times the zipper's signal and two holes 1.13–1.20 times; the zipper's

n	placement on the 20-cycle	two-qubit gates per layer	duration	λ (four classifiers)
18	two holes, moved every layer	22	6.7 μs	0.61–0.68
20	the ring itself	20	5.6 μs	0.78–0.86
22	two pendants	24	9.2 μs	0.75–0.84

Table 3: The rule on one 20-qubit cycle of the `ibm_kingston` day twin, $L = 16$ (standard errors of each λ 0.01–0.03). Predictions declared for $n = 20$ (≥ 0.61) and $n = 22$ (0.50–0.86) held.

output was also noisier (correlation with the noiseless output down to 0.89). The place matters as well: at $n = 18$ the zipper placed by PSF-Zero elsewhere on the device kept 0.63–0.66 with the same classifiers, 0.06–0.10 more than on this cycle.

n	rule's schedule	gates / μs	zipper on the same qubits: gates / μs	rule's λ	zipper's λ	verdict
18	two holes	352 / 6.7	656 / 10.6	0.61–0.68	0.54–0.59	WIN 4/4
20	the ring itself	320 / 5.6	736 / 10.6	0.78–0.86	0.44–0.60	WIN 4/4

Table 4: Only the schedule changed: same classifiers, inputs and qubits (`ibm_kingston` day twin, $L = 16$, four classifiers; standard errors 0.01–0.05).

6 Against Qiskit's search

Design. Qiskit (optimization level 2 on the day's Target, its default seed) ran on the same twin, classifiers and inputs as Table 2, with the declared rule of that table (the rule's and the zipper's rows were reused after a declared check that they reproduce exactly). Its circuits were simulated up to 24 active qubits; $n = 17$ was stopped for cost (about 3.5 h a classifier at 24 active qubits) and is reported as not run.

n	13	14	15	16	18
Qiskit's active qubits	20	20	24	16	22
two-qubit gates, rule / Qiskit	224 / 571	256 / 530	288 / 699	320 / 968	352 / 639
duration (μs), rule / Qiskit	9.2 / 53.7	9.2 / 49.1	9.3 / 65.5	9.3 / 94.7	6.7 / 53.0
rule's λ	0.76–0.81	0.73–0.82	0.78–0.81	0.77–0.80	0.61–0.68
Qiskit's λ	0.24–0.59	0.19–0.64	0.21–0.47	0.10–0.38	−0.05–0.28
rule vs Qiskit	WIN 4/4	WIN 4/4	WIN 4/4	WIN 4/4	WIN 4/4

Table 5: The rule against Qiskit on the `ibm_kingston` day twin, $L = 16$, four random classifiers. The zipper also kept more than Qiskit on every classifier. The prediction (WIN at every n run) held.

Reading. Qiskit's circuits spread a ring of 13–18 over 16–24 active qubits, with 1.8–3.0 times the rule's two-qubit gates and 5.3–10 times its duration. They kept −0.05–0.64 of the signal, below 0.3 on 14 of 20 classifiers, against 0.61–0.82 for the rule; where Qiskit's λ is positive the rule keeps 1.3–7.6 times as much, so a user needs 1.6–57 times fewer shots for the same precision. At $n = 18$ one classifier kept no measurable signal (-0.05 ± 0.26). Qiskit's circuit is the same for the four classifiers, yet its λ spans 0.19–0.64 at $n = 14$ and its correlation with the noiseless output is 0.28–0.90 (the rule's 0.99): its long circuits leave the output noisy as well as damped. Spreading the

ring over more qubits also widens the read-out's backward light cone, the only region whose noise reaches it [17]; we offer this as a qualitative reading, since an error product over that cone did not rank the schedules (Section 5). Beyond $n = 18$ Qiskit's circuits use 22–28 or more active qubits, which our trajectory simulation could not afford; this is a limit of our method, not a measurement of Qiskit. These are one setting of Qiskit on one day's twin with random classifiers; Paper 3 [2] compared Qiskit and the zipper through trained classifiers and on a device.

7 On devices

Design. With 18 s of free device time left, a two-arm comparison was rehearsed on the day twin first: the pendants' lead of about 0.1 would have been decided in at most about half the cases even with 1 minute of device time, because random classifiers at 16 layers spread their noiseless outputs over only 0.08 (s.d.). We therefore asked a narrower question: does the pendant schedule keep the signal its day's twin predicts? One arm ($n = 13$, $L = 16$, one random classifier, 64 random inputs, 250 shots), compiled and predicted for the Target fetched minutes before, AGREE if the device slope is within 2 standard errors of the twin's.

device	twin's λ (fixed before)	device λ	verdict	billed	zipper's twin λ (not run)
`ibm_kingston`	0.765	0.904 ± 0.124	AGREE	7 s	0.613
`ibm_fez`	0.591	0.504 ± 0.102	AGREE	7 s	0.432

Table 6: The pendant schedule on two devices, 26 October 2026 (224 two-qubit gates, 8–9 μs). The device correlations with the noiseless values (0.68, 0.53) are what 250 shots alone give at this spread (0.70, 0.60). On `ibm_fez` the twin was built from a calibration about 40 minutes older than the one under which the job ran.

Reading. On both devices the pendant schedule kept the signal its day's twin predicted, the lower value on `ibm_fez` following that day's lower prediction. These tests rule out a large loss (as Qiskit's circuits showed on `ibm_kingston` on 24 October [2]); they cannot see a difference of 0.1, and no zipper ran on a device beside them.

8 Limitations

The lower bound for pendants is proved for layered schedules on exactly the cycle and its leaves (a schedule that borrows nearby empty qubits is outside it) in a routing model that excludes measurement, feed-forward and ancilla-mediated gates, and checked by search only for small interleaved cases; the hole bound holds within a class; the zipper's is a search result. The signal results are on noise models of two calibrations and one device family, with random rather than trained classifiers, one placement per n, and four classifiers per cell; Qiskit ran at one optimization level with its default seed. The device tests are one job each, at a precision of about 0.1. Qiskit's structured swap strategies [9] were not among the arms. Count-optimal is not signal-optimal: duration and where the read-out sits also matter.

9 Reproducibility

- Compiler: PSF-Zero AI front end a26, `patches/psf_ai_compile_a26_2026-10-26/` (checks `check_a26.py`, `test_a26.py`).
- Runs, each with its declaration fixed by SHA-256 and time before the run (`prereg/drafts-2026-10-25/`, `prereg/drafts-2026-10-26/`, `prereg/drafts-2026-10-27/`): TOQB [3] `results/` `flow-d-2026-10-25`, and, all dated `2026-10-26`, `flow-e`, `flow-f`, `flow-g`, `flow-h`, `rule-big`, `esp-check`, `lam-rehearse`, `device-d1`, `device-d2`, `flow-i` (Qiskit), `flow-j-2026-10-27` (larger rings), `flow-k-2026-10-27` (same qubits, two schedules), and the searches and cycle counts of Sections 2 and 4, `lemma-check` (scripts and outputs; no declaration); the separate review of the pendant code, `pend-review-2026-10-26`.
- Device jobs: `dbfivq0m94ac738tldc0` (`ibm_kingston`) and `dbfjtvvsstms73fe0jc0` (`ibm_fez`); circuits, manifests and raw counts are in the two device folders.

Changes in version 1.3. The acknowledgment of IBM Quantum below. Version 1.2 (28 October 2026, released on GitHub only) reworded the related work and one sentence of the abstract and cited Kashif and Shafique. No number or verdict changes in either.

Acknowledgments

We acknowledge the use of IBM Quantum services for this work. The views expressed are those of the authors, and do not reflect the official policy or position of IBM or the IBM Quantum team. The device runs used the IBM Quantum Open Plan.

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